

Digital Logic Quiz

1

A classroom alarm should sound if **either** the fire sensor OR the smoke sensor is triggered.

Which logic gate best represents this behavior?

1. AND
 2. OR
 3. NOT
 4. XOR
-

2

A secure door opens only when **both** a keycard and a PIN code are correct.

Which logic gate best represents this behavior?

1. OR
 2. NOT
 3. AND
 4. NAND
-

3

A motion detector sends a signal of 1 when movement is detected. A system needs the opposite signal: 0 when movement is detected and 1 when there is no movement.

Which logic gate performs this operation?

1. OR
 2. AND
 3. XOR
 4. NOT
-

4

A computer backup should start only when:

- The laptop is plugged in, AND
- The user is not actively using it.

Which logic expression best matches this rule?

1. Power OR User Active
 2. Power AND NOT(User Active)
 3. NOT(Power) AND User Active
 4. NOT(Power OR User Active)
-

Answers

1.
B — An OR gate outputs 1 when either input is 1.
2.
C — An AND gate requires both conditions to be true.
3.
D — A NOT gate reverses the input signal.
4.
B — The backup runs only when power is available and the user is not active.